

AWAIS SHAH

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RESEARCH SUMMARY

Computer Science graduate with **one published and two under-review papers** in real-time object detection, edge–cloud inference, and XR adoption frameworks for resource-constrained embedded systems. Develops lightweight vision models (SSD MobileNet V2, YOLOv8) with a focus on **performance–efficiency trade-offs** on CPU-based and edge hardware. Proficient in Python-based ML workflows (TensorFlow, OpenCV). Interested in graduate research in **Efficient AI, Edge Intelligence, and Embedded Computer Vision**.

PUBLICATIONS

Shah, A. (2025). A Review of Real-Time Deep Learning–Based Object Detection Models for Resource-Constrained Embedded Systems. **[Peer-Reviewed]**

NUML International Journal of Engineering and Computing (NIJEC), 4(2), 13–29. [DOI: 10.52015/nijec.v4i2.110](#)

- Comprehensive survey benchmarking SSD MobileNet V2, YOLOv8, and peer architectures on latency, accuracy, and memory across embedded platforms.

Shah, A. (2025). Latency-Aware Edge–Cloud Perception Architecture for Latency-Sensitive Medical Metaverse Applications.

SSRN Preprint [Under Review] — ssrn.com/abstract=6254021

- Proposes hybrid edge–cloud inference via YOLOv8-tier detectors + Latency-Aware Decision Module (LDM) — achieves **83.6% latency reduction** vs. cloud-only while maintaining 69% detection agreement.
- Benchmarked across variable network conditions; validated real-time viability for medical XR applications.

Shah, A., & Khan, A. (2026). Development and Validation of the XR Readiness Index (XRI): Assessing XR Adoption Preparedness in Medical Education — Pakistan Case Study.

Zenodo Preprint — [DOI: 10.5281/zenodo.20026564](https://doi.org/10.5281/zenodo.20026564) (Shah: lead author; Khan: co-author)

- Developed and validated a computational framework for quantifying institutional XR readiness in LMIC medical education.
- Validated across **191 respondents** from all 7 provinces of Pakistan — mean XRI = 58.69/100; Cronbach's α = 0.750–0.801; KMO = 0.865.

EDUCATION

B.Sc. Computer Science — University of Haripur, Pakistan — 2020–2024 — CGPA: 3.14 / 4.00

Final Year Project: Real-Time License Plate Detection & Recognition System — fine-tuned SSD MobileNet V2 for multi-format plate localization and OCR-based text extraction on CPU hardware.

Relevant Coursework: Machine Learning · Computer Vision · Artificial Intelligence · Operating Systems · Data Structures & Algorithms

RESEARCH INTERESTS

- Real-time deep learning object detection on resource-constrained embedded and edge devices.
- Lightweight neural network architectures — quantization, pruning, and knowledge distillation.
- Performance–efficiency trade-offs for vision AI on low-power hardware (Raspberry Pi, Jetson Nano, ONNX).

RESEARCH & TECHNICAL PROJECTS

Real-Time License Plate Detection & Recognition System — [GitHub](#)

Python · TensorFlow · OpenCV · SSD MobileNet V2 · EasyOCR

- Fine-tuned SSD MobileNet V2 on 400+ annotated images for multi-format plate localization across varying lighting conditions.
- Applied 60% threshold-based OCR filtering to improve text extraction stability; UUID-based logging pipeline exports to CSV.
- Optimized CPU inference for parallel TensorFlow + PyTorch execution; deployed as prototype for parking automation and access-control.

Real-Time Sign Language Detection System — [GitHub](#)

Python · TensorFlow · OpenCV · SSD MobileNet V2

- Custom dataset of 280+ Pascal VOC–annotated images; fine-tuned SSD MobileNet V2 over 15,000 iterations.
- Achieved **AP@50: 0.990 · AP@75: 0.708 · mAP (0.50–0.95): 0.608** with real-time bounding-box visualization pipeline.

TECHNICAL SKILLS

Programming:	Python
Frameworks:	TensorFlow, PyTorch, OpenCV, Keras, EasyOCR
Deployment:	ONNX, Linux
Edge Hardware:	Raspberry Pi, Jetson Nano
Tools:	Git, Jupyter Notebook, Google Colab, VS Code, Labellmg
Libraries:	NumPy, Pandas, Matplotlib
Domains:	Edge AI, Embedded Vision, CNNs, Computer Vision
Languages:	Pashto (Native) · Urdu (Native) · English (Professional)

TEACHING & OUTREACH

Computer Science Teacher — Brainiac International School & College — Aug 2024 – Present

- Improved measurable comprehension for **53% of students** using the 5E Model, interactive examples, and project-based learning.
- Introduced AI modules covering machine learning fundamentals, Python, and real-world applications.

CERTIFICATIONS & ACHIEVEMENTS

- Python Programming — Kaggle (2025)
- Introduction to Machine Learning — Kaggle (2025)
- Guest Presenter, Annual IT Exhibition (2025) — ANPR system demo to GM of WAPDA; recognized for innovation and execution.